
HOW MUCH CAN A MACHINE UNDERSTAND TRANSIT DELAYS?

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ABSTRACT

Transit delays are a significant barrier to reliable public transportation in Toronto. In this study, we apply a range of machine learning techniques—including LSTM for sequence modeling, feedforward neural networks for regression, and XGBoost for classification—to analyze and predict TTC bus and subway delays. Using a hybrid fuzzy matching and Gemini-based pipeline, we standardized stop names across 80,000+ records for improved data quality. Spatial and temporal analyses revealed limited geographic clustering but highlighted strong delay patterns at terminal stations and during peak hours. Despite the LSTM model’s inability to capture meaningful temporal trends, the XGBoost classifier achieved high performance, with over 70% true positive rate in identifying delay occurrences. Weather features were found to have minimal predictive power. Our results suggest that classification-based approaches are most promising, and post-pandemic data (2022–2024) best reflect current TTC conditions for future modeling.

1 Introduction

1.1 Context

The Toronto Transit Commission (TTC) is Toronto’s public transit system, which assists over 484,381 passengers per Business Day [1]. Despite this, the TTC remains to be one of the most notorious social services of the city, with their reports revealing an urgent need for improvement with a reliability of only 58% [2]. Our project seeks to tackle this issue by using machine learning techniques to predict delays of TTC routes. With accurate predictions, riders will be able to make more informed decisions about their departure and arrival times, and with this data, the team is able to suggest optimal routes to riders. In addition, through the model, the team aims to understand the causes of delays, which opens up an opportunity for the TTC to lessen the delays.

1.2 Conditions of Success

The success of the model depends on various factors, such as the inherent correlation of the data used, the architecture used, and the optimization techniques and hyperparameters the team chooses. Ultimately, successful predictions involve maximizing the accuracy of the model to correctly predict presence of delays, which leads to more accurate departure times, and allows for minimized waiting (both at the departure and destination point) and transit time. More concretely, the model is successful if it yields more insights or better predictions than traditional methods of estimating delay occurrences with regression predictions.

2 Data

To have a comprehensive understanding, the team draws insights from various sources, such as bus, subway, and streetcar data.

2.1 Inputs

- TTC Bus Delay Data, provided as an open Toronto catalogue [3]
- TTC Subway Delay Data, provided as an open Toronto catalogue [3]

Date	Route	Time	Day	Location	Incident	Min Delay	Min Gap	Direction	Vehicle
1-Jan-24	89	02:08	Monday	KEELE AND GLENLAKE	Vision	10	20	N	7107
1-Jan-24	39	02:30	Monday	FINCH STATION	General Delay	20	40		8914
1-Jan-24	300	03:13	Monday	BLOOR AND MANNING	General Delay	0	0		8562
1-Jan-24	65	03:23	Monday	PARLIAMENT AND BLOOR	Security	0	0	N	8574
1-Jan-24	113	03:37	Monday	MAIN STATION	Security	0	0		8541
1-Jan-24	320	03:40	Monday	YONGE AND QUEENSQUAY	Operations - Operator	8	16	N	8192
1-Jan-24	171	04:00	Monday	MOUNT DENNIS GARAGE	General Delay	20	20	S	1402

Figure 1: Example of TTC delay data

2.2 Data Analysis Key Results

Our data analysis aims to identify critical factors influencing public transport delays, which guides the selection of relevant training data for the machine learning model. The key insights are as follows:

- **Regional Impact:** A heatmap visualizing delay patterns across different regions, highlighting areas with frequent or severe disruptions.
- **Temporal Trends:** Analysis of delay variations by time of day, day of the week, and long-term trends over past years to capture recurring patterns.
- **Frequent Reasons to Delay:** A histogram illustrating the most frequent reasons for delays.
- **Weather Influence:** Examination of weather conditions (e.g., heavy rainfall, snow, extreme temperatures) that correlate with delay frequency and severity.

2.3 Data Cleaning

2.3.1 Bus Data

Data Compilation and Standardization - The original bus data stored each month's data in a separate sheet and each year's in a different Excel spreadsheet. Additionally, the dataset recorded date and time as separate fields. To streamline processing, the spreadsheets were consolidated into a single CSV file, and the date and time columns were merged into a unified DateTime column. Outlier delay events exceeding 60 minutes were excluded to prevent skewing model performance, and all missing values were removed from the final merged dataset.

Stop Name Standardization - After this integration, the dataset contained approximately 80,000 unique bus stop entries, despite there being only around 9,000 unique stops in the TTC system. To standardize stop names, a combination of fuzzy matching and Gemini-based processing was employed.

Fuzzy Matching and Gemini Model - Fuzzy matching alone produced correct matches around 70% of the time, with an average confidence level of 86 (found through manual checking)—only slightly better than random. Gemini achieved high accuracy but was effective only for small sample sizes. Processing 100 or more entries at once on Gemini often led to errors and high computation time. To address these limitations, a hybrid approach was used. Fuzzy matching was combined with Gemini calls on smaller samples ($n = 10$) of data points with low fuzzy confidence to compare stop names in our dataset against those in the TTC Routes and Schedules

dataset[4]. Duplicate stops were removed, and in cases where Gemini assigned suboptimal matches, the mistakes were manually corrected.

2.3.2 Subway Data

Data Compilation and Standardization - To clean the subway data, the Excel spreadsheets were first combined using the same method as the bus data. To make the data easier to analyze, the TTC delay codes were then mapped to their respective descriptions using the TTC delay dataset[5]. Line numbers and station names were standardized using the fuzzy matching with the TTC Routes and Schedules dataset[4]. Notably, subway station names exhibited little spelling variations, allowing fuzzy matching alone to effectively correct discrepancies and resolve mismatched line numbers.

Handling Zero Data - After standardizing the dataset, of the approximately 200,000 remaining data points, around 135,000 had both "minutes delay" and "minutes gap" recorded as zero. The causes for this anomaly was investigated, revealing that the most common causes for these delays included speed control, ill or injured customers at the station, and operator overspeeding. These, along with other frequent causes, seemed unlikely to result in significant delays. Consequently, an intermediate dataset was saved with these entries, while a filtered version excluding these data points was created for further subway data analysis.

2.3.3 Weather Data

To investigate the relationship between weather conditions and subway delays, we integrated TTC delay logs with daily weather reports collected from Toronto City Centre [7]. Delay records were first aggregated by day, and both the mean delay duration as well as the total delay count, were computed. These values were then merged with daily weather features including temperature (average, minimum, and maximum), precipitation, wind speed, and wind direction.

3 Method

3.1 Location Analysis

To examine spatial trends in bus delays, each stop in the dataset was geocoded with its corresponding latitude and longitude. Data preprocessing included the removal of extreme delay values exceeding 60 minutes, which were considered outliers. Using the cleaned dataset, we generated contour heatmaps and 3D plots to visualize both the frequency and mean delay duration at each geographic location.

To improve visual clarity, particularly in areas with high stop density, we produced blurred versions of these maps by rounding coordinates to the nearest 0.01 degrees and annotated major transfer points such as Union Station, Finch Station, and Kennedy Station. Spatial clustering patterns were then analyzed to identify zones with consistently high delays.

Finally, delay patterns at specific locations were compared against historical records to assess the persistence of transit inefficiencies over time and to highlight stations with systematically poor performance.

3.2 Time Analysis

Exploratory Data Analysis: Using multiple time scales, including hour of the day, day of the week, and long-term year-over-year trends, we computed and visualized the mean delay at different hours of the day, both with and without outlier filtering. This enables investigation to the peak congestion periods and assess the consistency of delay patterns. Weekday and weekend delay distributions were compared to evaluate operational differences between regular service and reduced schedules.

To capture broader temporal dynamics, a 30-day rolling average of delay durations is computed, revealing seasonal variations and gradual trends over time.

Major disruptions such as the COVID-19 pandemic were identified and their impact on system performance quantified by observing sharp deviations from baseline delay levels.

Sequence-based LSTM model is utilized to extract temporal dependency of subway delays, using a 30-minute interval encoding of station-level delays, day-of-week, and time features. The model was trained using sliding windows on historical data from 2016–2024 and 2020–2024, and evaluated its ability to predict future delays from past temporal sequences.

3.3 Weather Data Analysis

We then trained a simple feedforward neural network to predict two targets: the average daily delay and the number of delay events, using six weather variables as input features. The network architecture included two hidden layers of 64 neurons each with ReLU activation, optimized using mean squared error loss and the Adam optimizer. The dataset was split chronologically into a training set and a 200-day test set. Learning rates were chosen to be $lr = 0.01$ based on validation set performance for $lr \in [0.000005, 0.01]$, with the range tested with a grid search of $n = 10$. Model performance was monitored using training loss and validation R^2 , and the learned relationships between weather conditions and delay patterns were further analyzed in the discussion section.

3.4 Binary Delay Prediction

Identification of flaws within the data and previously explored methods led to the development of a Binary Delay Classifier. In order to properly model this, a certain degree of data engineering was necessary. Since the raw dataset only contained positive (delayed) events, a synthetic dataset of non-delay records was generated based on the TTC’s scheduled train frequencies and operating hours [4]. Synthetic data was generated with both stochastic and non-stochastic methods, with performance varying as consequence. A record was engineered to include features selected for prediction, including hour, minute, day of the week, Station ID, line number, bound direction, ridership levels, rush hour indicator, and terminal station flag.

Logistic Regression was chosen as a baseline classifier to its computational simplicity and ease of interpretability. XGBoost was chosen to create our State-of-the-Art Classification Model due to its strong computational ability. The Classification Model was trained on fused synthetic and observed data, predicting the occurrence of delays at a subway station.

Due to the incremental nature of the synthetically generated data, there was a large class imbalance of 1 delay to every 1000 non-delays. Class imbalance was addressed using a two-step strategy: undersampling the majority (non-delay) class and applying SMOTE (Synthetic Minority Over-sampling Technique) to augment the minority (delay) class. Model performance was evaluated using accuracy, precision, recall, ROC AUC, and confusion matrices to assess both overall classification quality and the ability to detect rare delay events. Results are discussed in further details in section 4.4.

Interpretability was explored by computing feature importance with XGBoost’s built-in scoring method. This allowed for us to identify key predictive variables; detailed results are presented in the Discussion section.

4 Results & Discussion

4.1 Spatial Delay Patterns and Limitations

Results

Despite the use of high-resolution spatial data and comprehensive delay logs, our analysis revealed no consistent geographic patterns in TTC delay occurrences. As shown in Figure 2, delays were distributed relatively evenly across the city with no clear hotspots aside from major transfer stations like Union Station and Kennedy Station.

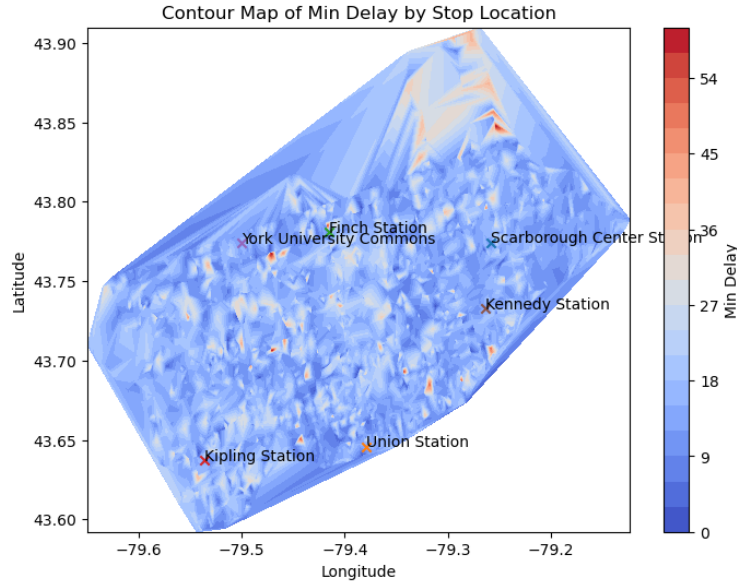


Figure 2: Contour heatmap of mean delay durations across TTC bus stop locations. High-traffic transfer stations such as Kennedy, Finch, and Kipling show elevated delays, but broader patterns remain unclear.

When blurred by rounding latitude and longitude to the nearest 0.01 degrees, the patterns remained similarly dispersed, suggesting a lack of strong local clustering.

Key Findings

1. Limitation with Data Collection - TTC logs each delay only at its first point of occurrence. Hence, when a train is delayed at a terminal station, downstream effects on other stations along the line are not recorded. This design introduces a fundamental blind spot: we do not observe the propagation of delay across the network, which makes it impossible to learn correlations or causal chains between stations. In modeling terms, this restricts our ability to capture spatial or sequential dependencies, because subsequent nodes (stations) may be affected in reality but are invisible in the dataset.

2. Terminal Stations & Geometric Correlations - A significant portion of recorded delays occur at terminal stations (e.g., Kennedy, Finch, Kipling), where trains originate or turn back. These stations are typically located at the geographic edges of the network and are spatially distant from each other. As a result, even if delays were to propagate, the terminal-centric nature of the delay records further reduces the chances of detecting spatial clustering. There's no natural neighborhood structure connecting these distant points, which undermines assumptions made by models like GNNs or spatial clustering techniques.

As a result, this spatial uniformity presents a challenge for spatial models like Graph Neural Networks (GNNs), which typically rely on structured node relationships and neighborhood-based propagation of information. In our case, the lack of spatial autocorrelation implies that the delay at one location offers little predictive power for delays at nearby stops, thus limiting the applicability of spatial deep learning methods in this context.

4.2 Temporal Pattern Extraction

LSTM's Inability to Capture Temporal Patterns

Despite extensive training using historical subway delay sequences from 2016 to 2024, the LSTM model failed to learn meaningful temporal patterns. The results can be referenced at 3. The cross-entropy loss remained constant across all epochs, and the model consistently

predicted the majority class—indicating no delay. Similar results were retrieved with Binary Cross Entropy with Logits when subway stop delays were instead evaluated as on-hot encoded arrays. This outcome highlights a key limitation of the dataset: delay events are both sparse and isolated, making it difficult to extract sequential dependencies using standard recurrent architectures. Additionally, since only one station is delayed at most per time step, the signal available for learning inter-station or temporal relationships is minimal. These findings suggest that richer features or a more granular representation of transit operations may be necessary to capture temporal dynamics effectively.

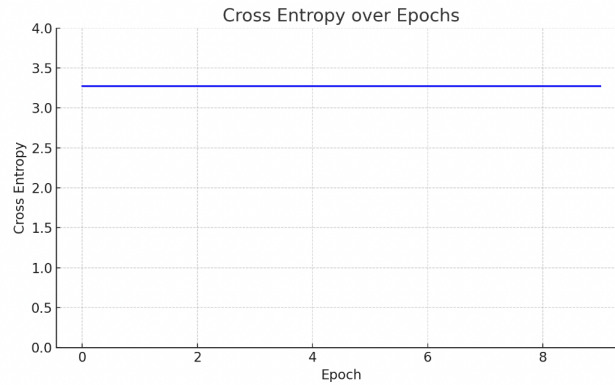


Figure 3: Cross-entropy loss over training epochs for the LSTM model. The flat curve indicates the model’s inability to learn meaningful temporal patterns, likely due to the limitations within dataset.

Effects of COVID Outbreak to TTC Delay

Figure 4 shows a noticeable increase in mean delay after the COVID-19 outbreak in early 2020. This outcome aligns with the financial and operational challenges reported in [6]. The TTC faced severe budget shortfalls, which led to service cuts, staffing shortages, and deferred infrastructure maintenance, all of which likely contributed to longer delays. Additionally, the decline in ridership and unpredictable fare revenue may have made it difficult to recover service levels, which further exacerbated the disruptions. This graph reveals that external events such as the pandemic can affect transit reliability. The team also wanted to use the delay by month

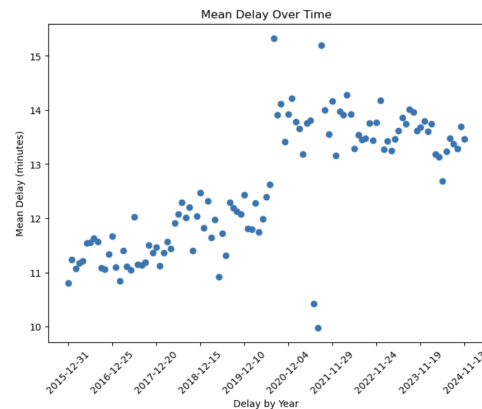


Figure 4: Bus Data Analysis, Mean Delay in minutes over time, plotted against delay by year.

data to see if weather plays a role in the bus delay. However, the result in figure 14 surprisingly reveals insights about the COVID-19 effects, instead of weather. We observe that the data from years 2016-2019, and years 2022-2024 are clustered. Years 2020-2021 seem to be outliers. This

indicates that for future analysis, we can put more weights on training data from 2022-2024, which are more recent and are more representative of current TTC delay trends.

4.3 Weather Data

Weather Data was incorporated both in a baseline model of Multiple Variate Linear Regression and a Feed Forward Neural Network. The feature was initially chosen due to the intuitive and correlative nature of weather and transit times, as determined by other transit organizations [8]. While the loss metric demonstrated successful model representation learning (see figure 5), the model was not able to properly represent variance within the data.

Both the baseline Linear Regression model and the Feed Forward Neural Network Model demonstrated extremely low R^2 values on validation sets (see table 1). Notably, best performing model, in terms of R^2 , resulted in a singular value (i.e. a means-based model) (see figure 6). This supports the lack of correlation between the weather statistics and delays, eliminating its viability as a predictive feature.

Model	Mean Delay R^2	Count of Delay R^2
Linear Regression	-0.0366	-0.3193
NN (lr = 0.01)	-0.0680	-0.1479
NN (lr = 0.00005)	-0.3744	-0.3983

Table 1: Comparison of R^2 scores for different models

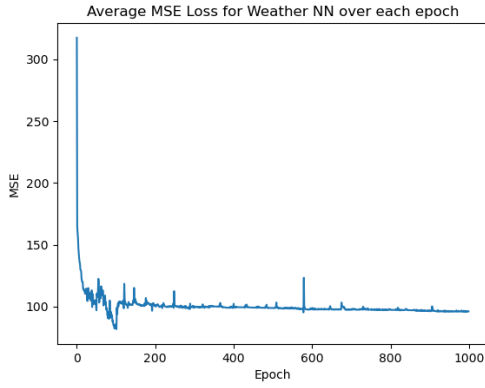


Figure 5: MSE Loss for Weather Data FFNN

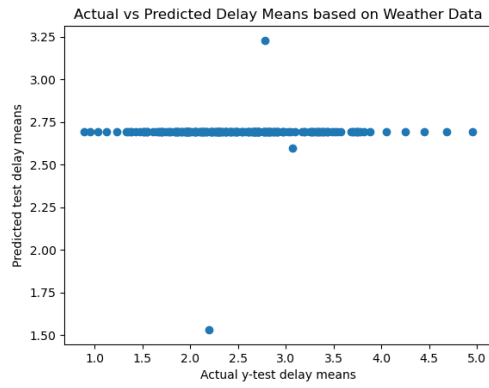


Figure 6: Results of FFNN with Weather

4.4 Binary Delay Predictability

Results

Logistic Regression Reveals poor results with 69.09% True Negative (No Delay) and 58.01% True Positive (Delay) classification rate. Binary Delay Classification with XGBoost results in exceptionally good results with 94.15% True Negative (No Delay) and 70.43% True Positive (Delay) classification rate. This reveals a distinct distribution between Delay and No-Delay selections, and **outperforms the baseline model**. Further, Feature Importance reveal important insight to the cause of Delay classifications, particularly features concerning time and ridership. We note that, comparatively, the location of the station (as determined by Station ID) and the direction of the train (Bound_ID) do not actively influence the selection of Delay vs Non-Delay.

Key Findings

The XGBoost model was able to successfully predict the occurrence of delays within the TTC transit system, beating the baseline Logistic regression model by 25%. Additionally, we were able to pinpoint key features that influence transit delays to time and ridership. These results provide key insights to the occurrence of delay; the ability for such explicit identification allow for actionable solutions towards current TTC delay issues.

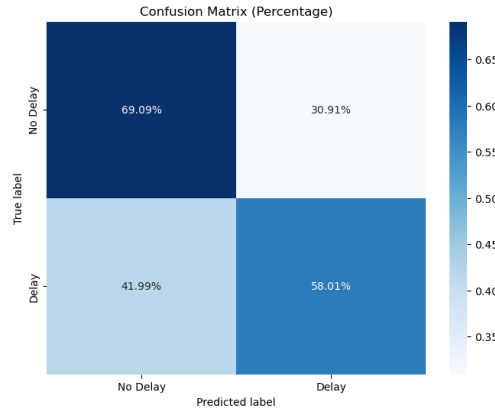


Figure 7: Results of Logistic Regression

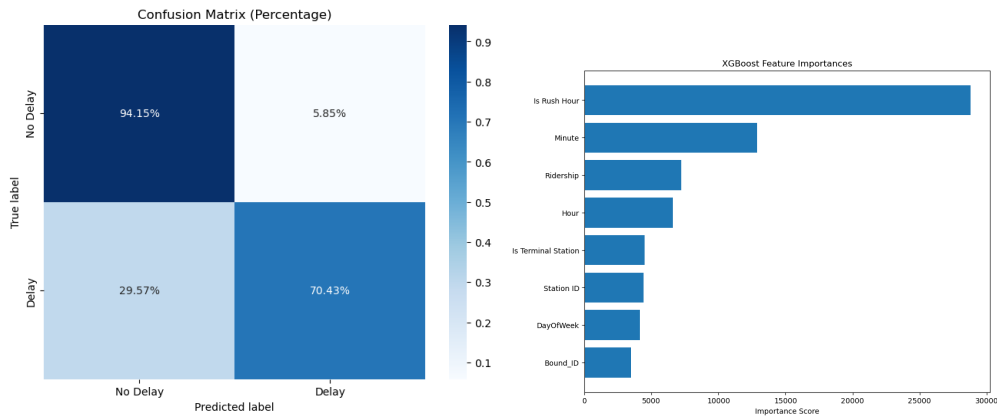


Figure 8: XGBoost Confusion Matrix

Figure 9: XGBoost Feature Importance

5 Conclusion & Reflection

This report details the analysis of TTC transit delays, with applications of Exploratory Data Analysis, Linear Regression, Feed Forward Neural Networks, LSTMs, Logistic Regression, and XGBoost Classifiers. Our Analysis reveals significant weakness within current data collection methods and the consequent inhibition of complex Machine Learning application towards specific station and delay time prediction. However, applications of classification methods, namely that with XGBoost, reveal successful distinction between features of delayed and undelayed trains. Inspection of feature influence within the classification models reveals that dominant features are time and ridership. While the inability to accurately predict the duration and place of transit delay proves a flaw towards passenger-friendly data understanding, classification processes demonstrated in our report may allow significant advancements in transit systems around the world.

References

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- [2] D'Souza, A.: Transit report cards reveal an urgent need for improvement in the Toronto region. *Toronto Region Board of Trade*, <https://bot.com/News/Transit-Report-Cards-Reveal-an-Urgent-Need-for-Improvement-in-the-Toronto-Region>, accessed Jan. 31, 2025.
- [3] Toronto Transit Commission: About TTC Bus Delay Data. *Toronto*, <https://open.toronto.ca/dataset/ttc-bus-delay-data/>, Dec. 17, 2024.
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- [6] PressProgress: Funding cuts have left Toronto with a \$125 million “revenue shortfall” on public transit. *PressProgress*, <https://pressprogress.ca/funding-cuts-have-left-toronto-with-a-125-million-revenue-shortfall-on-public-transit/>, June 2023.
- [7] The Weather’s Record Keeper, *Meteostat*, <https://meteostat.net/en/>, accessed Apr. 7, 2025.
- [8] Winter safety tips for transit riders, *Niagara Region Transit*, <https://www.nrtransit.ca/news/winter-safety-tips-for-transit-riders/>, accessed Apr. 7, 2025.

A Appendix / supplemental material

Optionally include supplemental material (complete proofs, additional experiments and plots) in appendix. All such materials **SHOULD** be included in the main submission.

A.1 GitHub URL

<https://github.com/michellyucollins/TardyTerminatorCommittee>

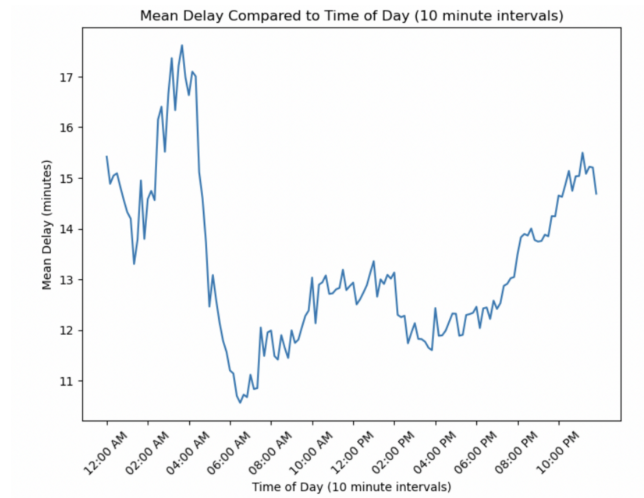


Figure 10: TTC but delay data by the time of the day. This does not reveal too much correlation between peak hours and delays. The delays at midnight is likely due to reduced schedules or staff.

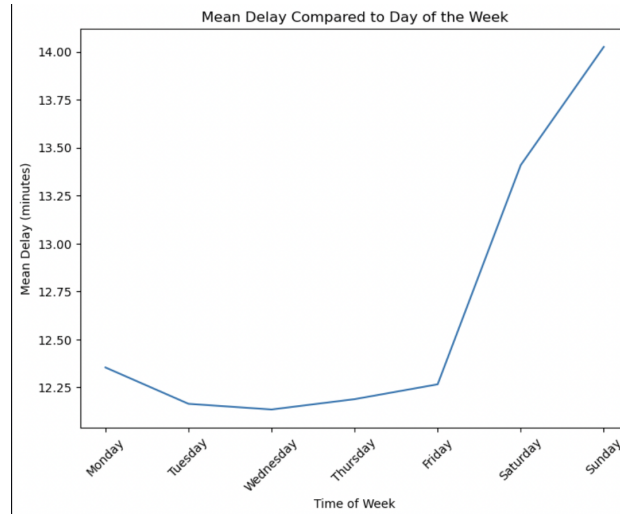


Figure 11: TTC bus delay data by the time of the week. The higher mean delay over the weekend is likely due to reduced staff - additional sources are needed to solidify this argument.

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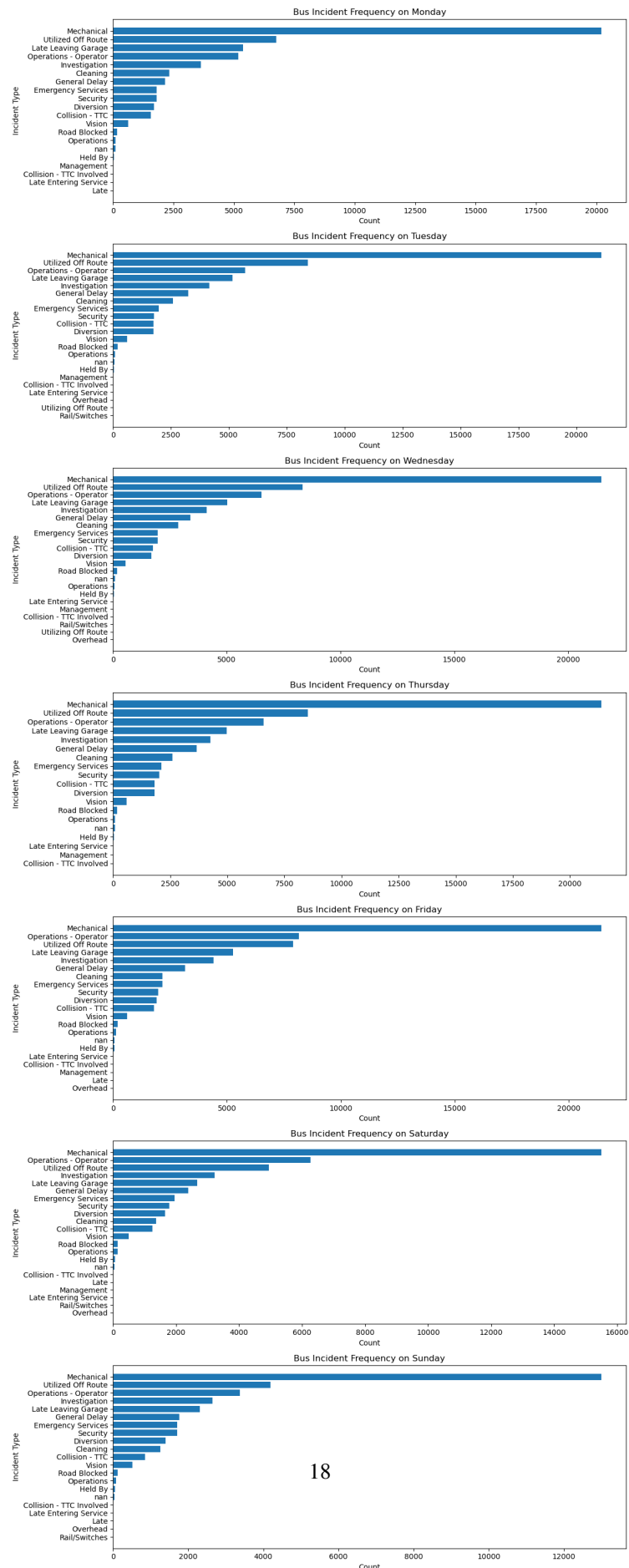


Figure 12: TTC subway delay data by the time of the week. In contrast to figure 11, the frequency over the weekend is actually the lowest. This means that frequency of delays during the weekend delays are less than weekdays, however, the amount of time in delay is actually longer. This might indicate longer time to resolve delay issues over the weekend.

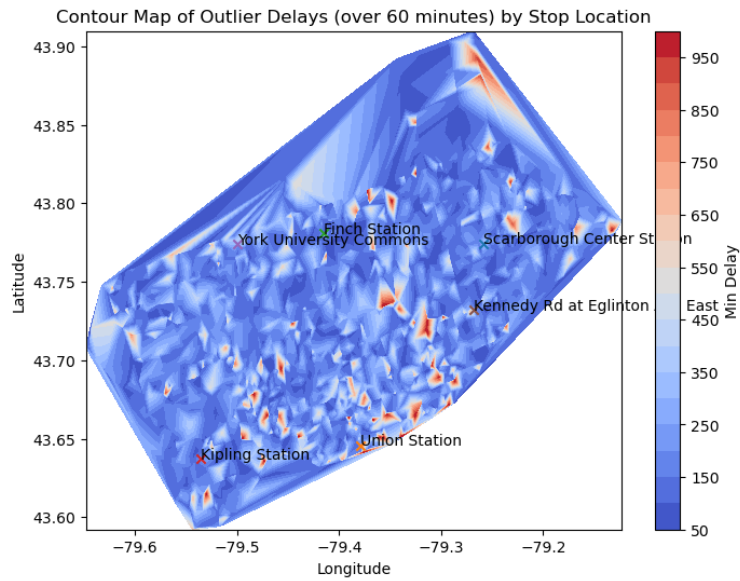


Figure 13: TTC bus delay data.

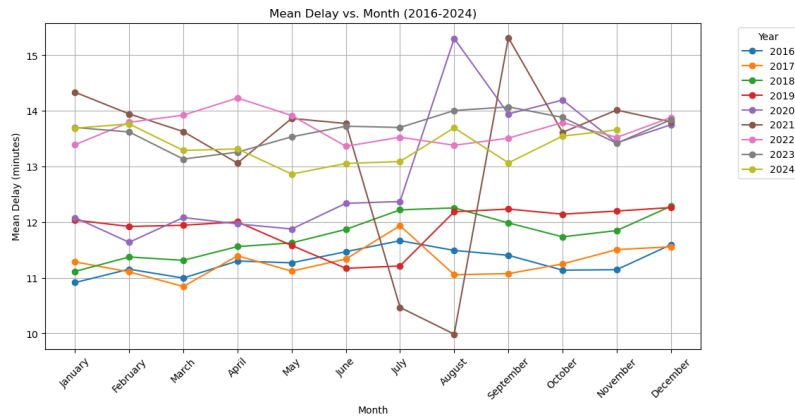


Figure 14: Mean Delay Delay by Month, Plotted across years from 2016-2024. Clustering can be seen for years 2016-2010, and 2022-2024; the latter two years exhibit a notable surge in mean delay time. The years of 2020-2021 are rather random.